Public Dissemination Materials Overview

Machine Learning for Seakeeping Control in Waves

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1. Purpose and Public Value

This document introduces the research materials made available by Sungshin Women's University for publication on the research centre's website. The materials support understanding, reproduction, and further investigation of data-driven ship-motion forecasting. The public GitHub repository is the principal access point and version record [1]. It brings together source code, ship-motion datasets, an explanatory notebook, and documentation covering both theoretical foundations and practical use.

The principal documented task is to predict future six-degree-of-freedom (6-DOF) ship motions—surge, sway, heave, roll, pitch, and yaw—from an observed motion history. Two complementary implementations, a bidirectional LSTM (BiLSTM) and a causal ModernTCN-style convolutional model, are provided with a shared experimental pipeline. The materials are intended for researchers, postgraduate students, educators, and engineers investigating time-series prediction and seakeeping-related analysis.

2. Materials and File Formats

Material	Purpose and contents	Format / access
Public repository	Versioned entry point, project description, directory structure, and links to all release materials.	GitHub; `README.md` [1]
Final training implementations	`py_BiLSTM_std.py` and `py_modernTCN_std.py`: loading, preprocessing, model construction, hyperparameter search, training, evaluation, and plotting.	Python source (`.py`) [1]
Detailed code review and user manual	Function and tensor explanations, installation, data preparation, training, interruption/resumption, inference, troubleshooting, and improvement recommendations. English and Korean editions.	Markdown and PDF; English Word edition [2]
Theory and previous experiments	`LRF-SSWU-CodeDocumentation_Theory.pdf` : forecasting formulation, architecture diagrams, normalization, optimization, search spaces, metrics, and historical results.	PDF, 9 pages [3]
Supporting research assets	`dataset/`, `dataset_grid/`, `ComrisonOfModels.ipynb`, and an earlier Causal ModernTCN analysis report.	CSV/OUT data, Jupyter notebook, Markdown [1]

Together, the theory document explains the scientific design, the manual explains how the implementation behaves and how to use it, and the repository supplies the files needed to inspect and conduct experiments. This overview is supplied in PDF and editable Word formats for direct use by the research centre.

3. Scientific and Technical Content

Prediction problem. Both final scripts use six motion channels jointly. Their default configuration takes 128 samples of history and predicts 128 future samples in one forward pass. At the assumed 2 Hz sampling rate, future lead times run from 0.5 to 64 seconds after the last observation; window starts advance by 16 samples. Translations are represented in metres and rotations in degrees.

Complementary architectures. The BiLSTM encodes the observed window in both directions and concatenates the last layer's final forward/backward states before a linear forecast head. Its backward pass reads only already observed history. The causal ModernTCN implementation uses left-padded depthwise temporal convolutions, per-time-step channel normalization, pointwise channel mixing, residual connections, and a temporal projection head. The head uses the complete observed window. It is the project's specific causal implementation, rather than a claim of exact reproduction of the general ModernTCN architecture [4,5].

Common experimental procedure. Files are split into training, validation, and test sets before windows are generated. Per-channel z-score statistics are fitted to training data only. Optuna explores architecture and training settings using validation MSE, TPE sampling, and median pruning [6]. Defaults allow 40 trials with up to 40 epochs each. A new model is then initialized with the selected settings and trained for up to 300 epochs on the original training split, retaining validation for checkpoint selection and early stopping. AdamW, learning-rate reduction, and gradient norm clipping are used. Test reporting includes standardized MSE/MAE and per-DOF RMSE in physical units.

4. Using the Materials

1. **Read and obtain files.** Start with the repository README, then consult the English user manual [2] and the theory document [3]. Clone or download the repository; the Korean manual provides an additional reading option.
2. **Prepare compatible data.** Create motion CSVs with `x1`–`x6` in the documented order, finite values, and a verified 0.5-second sampling interval. Each training file requires at least 256 rows. Set `ID_DATA_DIR` or prepare `csv_dataset/`. Repository data folders can have different sampling intervals and exported column labels; validate and preprocess each source before use. `Wave_` files are excluded by the final scripts.
3. **Set up and run.** Follow the manual's verified Python/PyTorch dependency guidance, preflight checks, and reduced synthetic example before a full experiment. CPU execution is supported; CUDA is selected when available. Give each experiment a distinct output path.
4. **Inspect and reuse outputs.** Preserve the best model weights, scaler and split lists, hyperparameters, SQLite trial database, training history, metrics, and prediction plots together. The manual includes a saved-model inference example. Final trained weights are generated by running the pipeline and are not included in this documented release.

Interpretation and validation. The manual records synthetic end-to-end checks and identifies implementation issues, including equal weighting of unequal evaluation batches and ModernTCN's handling of historical interruption flags. Proposed fixes are documented but have not been applied to the final source scripts. Historical scores in the theory PDF are separate from those checks; the full real-data search has not been newly reproduced during documentation preparation. These scripts provide point forecasts, without implemented uncertainty intervals or a validated operational control system.

5. Access, Website Publication, and Version Identification

The repository README is the recommended landing page [1]. It links directly to the two final training scripts, both language editions of the user manual, the theory PDF, and this overview. Markdown files support browser reading and version comparison; PDF files support stable reading and printing; the English Word files support editorial preparation.

For research-centre publication, the accompanying files are organised as follows:

Publication item	File or location
Overview for the website	`LRF_SSWU_Public_Dissemination_Overview.pdf` and the corresponding `.docx`
Theory explanation	`LRF-SSWU-CodeDocumentation_Theory.pdf`
English user documentation	`BiLSTM_ModernTCN_Code_Review_and_User_Manual.md`, with corresponding `.pdf` and `.docx`
Korean user documentation	`BiLSTM_ModernTCN_Code_Review_and_User_Manual-kor.md`, with corresponding `.pdf`
Training source and supporting datasets/notebook	Repository download or direct links through the README [1]

The overview and principal explanatory PDFs can be hosted directly on the research-centre website, with a link to GitHub for source code, data, and subsequent revisions. To make a reported experiment traceable, record the repository commit, actual data files and preprocessing, split lists, settings, and software environment. The final training-code baseline reviewed in the manual is commit `31988e994fe6d35e1308760700ecbd2e4f3ec647`; the repository history identifies later documentation updates. Users citing the materials should include the institution, document title, repository URL, and the revision or access date used. Third-party dependencies retain their own attribution and licensing information.

6. References and Further Reading

[1] Sungshin Women's University. *Machine Learning for Seakeeping Control in Waves: Public Research Repository.* Project information, source code, datasets, notebook, and release index. https://github.com/yangjunahn/lrf-sswu

[2] *Detailed Code Review and User Manual for BiLSTM and Causal ModernTCN.* English edition, 19 October 2026; Korean translation and downloadable formats linked from the README.
English manual on GitHub

[3] *Causal ModernTCN and Bi-LSTM for 6-DOF Ship Motion Time Series Prediction and Hyperparameter Optimization: A Code Analysis Report.* Theory and prior experiments, 9 pages.
Theory PDF on GitHub

[4] Hochreiter, S., and Schmidhuber, J. (1997). Long Short-Term Memory. *Neural Computation*, 9(8), 1735–1780. doi:10.1162/neco.1997.9.8.1735.

[5] Luo, D., and Wang, X. (2024). ModernTCN: A Modern Pure Convolution Structure for General Time Series Analysis. *ICLR 2024*. OpenReview.

[6] Akiba, T., et al. (2019). Optuna: A Next-generation Hyperparameter Optimization Framework. *KDD 2019*. Author preprint, arXiv:1907.10902.

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